
How much can a Scrabble move score?

Exact records from one tile to seven

Abstract. How many points can one Scrabble move score if it adds exactly k tiles? We answer this question for $k = 1, \dots, 7$ under NWL23, the North American tournament dictionary, and CSW24, the international Collins dictionary. The answers range from 213 points for one tile to 1787 for seven. The proof separates three questions: can the chosen words be scored together, can the surrounding board be legal, and can that board arise in an actual game? Elementary bounds reduce the first question to finite enumeration. Completion arguments settle the few remaining boards, and explicit move histories settle the third. The larger dictionary usually gives the larger record, but not for six tiles.

Keywords: Scrabble, extremal combinatorics, computer-assisted proof, dictionary comparison.

1. A TWO-TILE SURPRISE The play that prompted this article looks almost modest. We label the columns A–O from left to right and the rows 1–15 from top to bottom. A still from the round-17 broadcast of the 2026 Word Cup shows Evan Chester facing Steve Grob, both listed in the event’s top division [1]. The letters AG were already side by side near the right edge. Chester put Z on L8 and S on O8, around the old letters on M8 and N8, making ZAGS. The Z landed on a double-letter square. The S landed on a triple-word square and also completed the vertical word VIDIOTS. Those two little tiles scored as follows. A Z is worth 10 points, doubled here to 20; A, G, and S are worth 1, 2, and 1.

$$3(20 + 1 + 2 + 1) + 3(4 + 1 + 2 + 1 + 1 + 1 + 1) = 72 + 33 = 105$$

points. The broadcast called it a competition record for a two-tile play. That historical claim is only our starting point; none of the proofs below depends on it.

We recall the rules we need. The board is a 15×15 grid, and H8 is its centre. The bag contains 98 letter tiles of fixed multiplicities and two blanks, and a player holds at most seven tiles. The first move must cover H8. Later, a player may pass or place one to seven tiles in a single row or column, possibly with old tiles between them; every square from the first tile to the last must then be occupied, forming one contiguous main word. The occupied board must remain connected. Every maximal horizontal or vertical string of length at least two that the move creates must belong to the chosen dictionary [2].

Each letter has a face value. Some squares double or triple the value of a new tile; others double or triple an entire word containing that tile. A premium works only when its square is first covered. All words made by the move are scored: the word along the direction of play is the *main word*, and the perpendicular words through new tiles are *crosswords*. Thus ZAGS was the main word and VIDIOTS a crossword. A blank may represent any letter but always scores zero. Playing all seven rack tiles adds a 50-point bonus. The full tile list and premium-square coordinates are recorded in the Supplementary Material.

This makes Chester’s 105 points less mysterious, but it raises a cleaner question. Fix k , the number of tiles placed on the turn. How large can the score be if the players

and the draws cooperate perfectly? Write this maximum as $M_k(D)$, where D is the permitted dictionary. We use NWL23, the North American tournament dictionary, and CSW24, the international Collins dictionary [3, 4, 5]. The dictionary is part of the rules: a word accepted in one form of the game may fail in the other.

Real tournament records provide useful perspective. NASPA lists Karl Khoshnaw's 392-point CAZIKUES, played in Manchester in 1982, and three 311-point plays in its North American tournament category [6]. Constructed positions go much higher. Scrabulizer records 1780- and 1785-point constructions from 2008, and the contributor DejMar reported a 1787-point CSW15 construction in 2016 [7, 8]. In 2024 Bob Lucassen found the 1786-point NWL23 construction used below [9]. Our concern is not priority for those numbers, but proof: are they really the best possible for their stated dictionaries, and what happens for every smaller value of k ?

Theorem 1 (The fourteen records). *For ordinary move scoring on the standard English board,*

k	1	2	3	4	5	6	7
$M_k(\text{NWL23})$	213	459	882	1130	1401	1723	1786
$M_k(\text{CSW24})$	225	483	912	1349	1478	1721	1787

Every value is attained in a legal game.

We exclude challenge bonuses, time penalties, endgame rack transfers, and unchallenged invalid words. The theorem is existential: the positions need not be sensible tournament strategy. The two finite word sets are versioned by counts and SHA-256 hashes in the Supplementary Material, making the statement independent of later dictionary revisions.

2. WHAT AN UPPER BOUND MUST REMEMBER A candidate record has three layers. First come the words scored on the final turn. They must share letters without asking for more tiles than the bag contains. Second comes the rest of the board: it must be connected and contain no stray nonwords. Third comes time: the board must be constructible from an opening word through legal turns. For an upper bound we may safely forget some of these restrictions; for equality we must restore all of them.

Suppose first that $k \geq 2$. By reflecting across the main diagonal, we may take the move to be horizontal. It makes one horizontal main word W and at most k vertical crosswords, one through each new tile. A one-tile move makes at most two words and is treated in both orientations.

Lemma 2 (Deletion fragments). *Delete the newly placed tiles from a word scored on the final turn. Every remaining consecutive fragment of length at least two was already a word.*

Proof. The letters of such a fragment were already present. An internal fragment ends where the board was empty before the move. An exterior fragment cannot extend beyond the final word, which was chosen maximal. Thus every nontrivial fragment was a maximal string before the turn. A one-letter or empty fragment imposes no condition: the single old tile may have been connected in the other direction. ■

Lemma 3 (Premium ceiling). *The main-word multiplier of a k -tile move is at most $3^{\min(k,3)}$. A crossword multiplier is at most three.*

Proof. An edge row contains three triple-word squares and no double-word squares. The middle row contains two triple-word squares and one double-word square, whose

product is only 18. Every other row contains at most two double-word squares and no triple-word square. At most k of these premiums can be activated. Transposition gives the same statement for columns. A crossword contains only one new square, so it activates at most one word premium. ■

This bound concerns multipliers, not the whole score. To keep the remaining arithmetic exact, let $v(a)$ be the face value of the letter a , and let λ_p and μ_p be the letter and word multipliers at square p . If I indexes the new letters of W , then the all-real-tile score of the main word is

$$A(W, I) = \left(\prod_{i \in I} \mu_{p_i} \right) \left[\sum_{i=1}^{|W|} v(W_i) + \sum_{i \in I} (\lambda_{p_i} - 1) v(W_i) \right]. \quad (1)$$

Old premium squares do not appear: they never reactivate.

For a fitted main word P , let $C_j(P)$ be the largest all-real score of any deletion-valid crossword through its j th new tile, including the option of no crossword. Then

$$U(P) = A(P) + \sum_{j=1}^k C_j(P) + 50 \mathbf{1}_{\{k=7\}} \quad (2)$$

is an upper bound. It gives each cross its favorite surrounding letters independently, even when those choices compete for the same physical tiles. That deliberate generosity makes $U(P)$ useful for discarding most placements.

The candidate patterns must also respect the bag. A crossing square is one physical tile even though it contributes to two words. For each scored square s , add its coefficients in all scored words and call the sum c_s . If the board displays n_a copies of a but the bag has only B_a real copies, at least $e_a = (n_a - B_a)_+$ of them are blanks. Order the coefficients of the a -squares as $c_{a,1} \leq \dots \leq c_{a,n_a}$.

Lemma 4 (Exact blank penalty). *The scored skeleton is impossible if $\sum_a e_a > 2$. Otherwise its greatest score on the scored squares under the tile supply is its all-real score minus*

$$\sum_a v(a) \sum_{j=1}^{e_a} c_{a,j}. \quad (3)$$

Proof. A blank on an a -square s loses exactly $v(a)c_s$. If a blank occupies a larger-coefficient square while a real a occupies a smaller one, swapping the two cannot lower the score. Repeating the swap places the forced blanks on the least expensive squares. The real-tile supplies are independent, and the two blanks are their only shared resource. ■

For the largest cases, a second upper bound keeps the Cartesian products small. Give each letter a any nonnegative rational price p_a . For an option o in crossword slot j , let s_{jo} be its score and a_{jo} its vector of old tile counts, excluding the shared new square. Let m count each physical main-word square once, let B be the real bag, and put

$$z_j = \max_o (s_{jo} - p \cdot a_{jo}), \quad t = \max(0, \max_a (p_a - v(a))).$$

Lemma 5 (Resource prices). *Every product through that main placement has pre-bingo score at most*

$$A + p \cdot (B - m) + \sum_j z_j + 2t.$$

For any options already chosen, their nonnegative reduced costs $z_j - s_{jo} + p \cdot a_{jo}$ may be subtracted from this bound.

Proof. If e_a blanks represent letter a , feasibility gives $m + \sum_j a_{jo_j} \leq B + e$ coordinatewise and $\sum_a e_a \leq 2$. Each blank loses at least $v(a)$. Hence the score is at most

$$A + \sum_j (s_{jo_j} - p \cdot a_{jo_j}) + p \cdot (B - m) + \sum_a (p_a - v(a))e_a.$$

The last sum is at most $2t$, and every unchosen option is bounded by z_j . ■

The prices used in the certificates are multiples of $1/1024$, so every comparison is an integer comparison. How the prices were found is irrelevant to validity: any non-negative prices satisfying the displayed inequalities give the bound.

We now make the finite reduction precise. A *scoring skeleton* is a triple (N, F, E) . Here N is the set of new squares, F assigns letters to the union of the main word and its crosswords, and E consists of squares forced empty: the squares just beyond those maximal words, and both perpendicular neighbors of a new tile that makes no crossword. Shared squares occur only once in F . A *completion* extends F to a final board, leaves E empty, uses the common 100-tile supply with at most two blanks, is connected, and has every maximal string of length at least two in D . If the skeleton contains old tiles, its move is not the opening move, so its completion must also contain H8. Every legal move has a completion; the converse need not hold.

The Chester play gives a small example. Its new set is $N = \{L8, O8\}$. The map F spells ZAGS on L8–O8 and VIDIOTS on O2–O8, counting O8 only once. The forced-empty set contains K8, O1, O9, L7, and L9: these prevent either scored word from extending and prevent the new Z from making an unlisted crossword. Deleting the new letters leaves the old words AG and VIDIOT. This skeleton remembers exactly what the final score forces and nothing else.

Lemma 6 (Line relaxation). *For one row, write a forced letter as itself, a forced empty square as #, and every other square as a dot. A left-to-right recursion decides exactly whether the dots can be filled or left empty so that each maximal occupied interval is either one square or a word in D . Every completion of a scoring skeleton passes this test in every row and column.*

Proof. At the first unprocessed square, either leave an optional dot empty, or choose each dictionary interval beginning there that matches all forced letters and avoids every #; continue after a separating empty square. These are exactly the possibilities for the first maximal interval. Induction on the unprocessed suffix proves the first statement. A true completion supplies one such filling. Allowing isolated squares and allowing different lines to reuse the same remaining tiles only enlarge the set of completions. ■

Lemma 7 (Hooks and reduced boards). *Suppose an old top-edge letter or block has only new or forced-empty horizontal neighbors. Every completion contains a perpendicular dictionary word through at least one of its squares. After one such mandatory hook is chosen for each block, every completion contains a reduced board obtained by*

retaining only the dictionary intervals needed to repair its forced nonwords. If no such reduced board respects one shared tile supply, no completion exists.

Proof. Before the final move, the block cannot connect through the new squares or through the outside of the board. Since the old board was connected, it must connect downward through a maximal perpendicular word. For each forced illegal run in a row, a completion has a unique maximal horizontal dictionary word containing that run; retain its squares and discard every optional square not so selected. This is one of the enumerated reduced boards and is a subboard of the completion. Finally, every retained square is one physical tile. Enumerating all hook and repair choices while debiting their union from a single multiset therefore rejects only tile-infeasible boards. ■

Proposition 8 (Completeness of the finite reduction). *Fix D , k , and a pre-bingo threshold q . The following enumeration contains the scoring skeleton of every legal k -tile move scoring at least q before the seven-tile bonus.*

1. *Choose a word $W \in D$, the k new positions in W , an orientation, and a fit on the board; retain only deletion-valid choices.*
2. *Retain a fit if its independent bound (2), without the bonus, is at least q . The resource-price bound may reject it only when the integer inequality in Lemma 5 is below q .*
3. *Enumerate the Cartesian product of all deletion-valid crossword options, merge shared squares, and apply the exact blank penalty.*
4. *Apply centre occupancy, line compatibility, mandatory-hook, reduced-board, and shared-bag tests as needed.*

Consequently, if every retained skeleton fails one of these necessary tests, no such move exists.

Proof. Take a legal move at or above q . Reflection lets us make its main word horizontal when $k \geq 2$. Its actual main word, new positions, orientation, and location occur in step 1 by Lemma 2. Its actual crosswords are among the independently chosen maxima, so it cannot fail step 2. They occur as one tuple in step 3; merging squares gives its physical multiset, and Lemma 4 gives the best possible placement of any forced blanks. Its final board is a completion of the resulting skeleton. Hence it passes the centre and compatible-line conditions, contains one of the enumerated words for each mandatory hook, contains the corresponding reduced board, and obeys the shared bag. Thus no stated rejection can remove the chosen move. ■

Table 1 is the complete audit trail. For $k \geq 2$, the column headed q omits the 50-point bonus when $k = 7$; “first stage” counts main placements left by the independent bound, and “exact stage” counts their shared-square, bag-feasible crossword tuples. For $k = 1$, those columns instead give deletion-valid word occurrences and crossing pairs scored above the live branch bound. A zero in the last column is the conclusion of a necessary test, not a failure to find an example.

The certificate of schema upper-bound-audit-v1 maps every row to its upper scan, exact-product output, final-completion output, and their hashes. Its verifier parses those artifacts and checks every count in the table.

Finally, an arrangement of valid words is not yet a game. The small boards use the following simple observation; the nearly full seven-tile boards receive explicit two-player rack certificates.

Lemma 9 (Sparse histories). *A legal positive-scoring placement history using at most 86 physical tiles can be realized by two players with legal racks and draws.*

Table 1. Certificate counts for the finite reduction.

D	k	q	First stage	Exact stage	Final necessary test
NWL23	1	–	240,528	36,941	largest score 213
CSW24	1	–	387,172	27,520	largest score 225
NWL23	2	459	81	1	largest product 459
CSW24	2	459	256	369	largest product 483
NWL23	3	883	4	0	product enumeration: 0
CSW24	3	883	7	11,350	largest product 912
NWL23	4	1131	10	0	product enumeration: 0
CSW24	4	1350	3	0	resource/product bounds: 0
NWL23	5	1402	2	0	product enumeration: 0
CSW24	5	1479	12	10	H8 occupancy: 0
NWL23	6	1724	9	20	rows leave 16; hooks: 0
CSW24	6	1722	21	15,809	rows leave 3,352; hooks leave 7; shared bag: 0
NWL23	7	1737	87	3,748	rows leave 255; hooks leave 34; 193 reduced boards; connectors: 0
CSW24	7	1738	168	420,538	rows leave 11,753; hooks leave 22; H8 leaves 9; shared bag: 0

Proof. Reserve for the second player seven tiles unused by the history, and let that player pass. Order the first player's required tiles by move, append seven unused tiles, deal the first seven, and draw along this list. Before each move the rack contains every tile that move requires. At most $86 + 7 = 93$ stream tiles are needed after the opponent's rack is reserved. Because every placement scores, the intervening single passes do not end the game; the appended filler tiles also prevent the builder from emptying a rack before the record move. Any tiles not yet named are placed at the inaccessible end of the bag. ■

3. ONE AND TWO TILES For a square p and a letter a , let $H(p, a)$ be the set of all deletion-valid occurrences of dictionary words that fit horizontally through p with distinguished letter a at p . Define $V(p, a)$ vertically. These sets are finite and explicit.

Proposition 10. $M_1(\text{NWL23}) = 213$ and $M_1(\text{CSW24}) = 225$.

Proof. A one-tile move makes at most one horizontal and one vertical word. By Lemma 2, it therefore appears in

$$\bigcup_{p,a} (H(p, a) \cup V(p, a) \cup (H(p, a) \times V(p, a))).$$

For every member of this union, the enumeration applies the premium at p , counts the common new square once in the tile supply but once in each word's score, and applies Lemma 4. This gives maxima 213 and 225.

For completeness, the implementation sorts each $H(p, a)$ and $V(p, a)$ by its all-real contribution. During the nested scan, the first unvisited term bounds every later term. A suffix is skipped only when that bound is smaller than the incumbent; induction over the sorted lists shows that no skipped pair can improve it. This branch-and-bound changes the running time, not the set being maximized.

Equality is simple. In NWL23, adding S at O15 completes HYDROXYZINES and MICROTECHNIQUES; both meet the new triple-word square, giving $3(38 + 33) = 213$. In CSW24, the same geometry completes METHOXYBENZENES and QUICKSILVERINGS, giving $3(41 + 34) = 225$. ■

Proposition 11. $M_2(\text{NWL23}) = 459$ and $M_2(\text{CSW24}) = 483$.

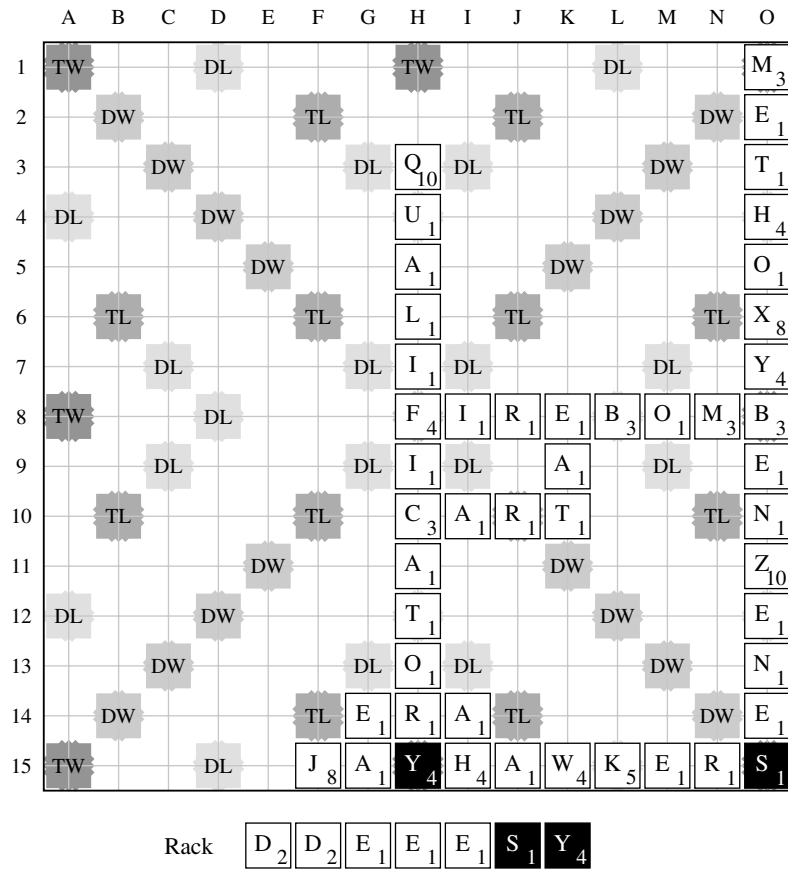


Figure 1. The 483-point CSW24 two-tile move. Black tiles are newly placed; the complete rack appears below the board.

Proof. Apply Proposition 8 with $q = 459$. The two $k = 2$ rows of Table 1 show that the largest exact scoring products are 459 and 483. These are upper bounds even though the product stage ignores connectivity and chronology.

For equality in NWL23, play Y at H15 and S at O15 to make PAYCHEQUES. The main word covers two triple-word squares; the crosses are XYLOGRAPHY and KATZENJAMMERS. Thus

$$9 \cdot 29 + 3 \cdot 29 + 3 \cdot 37 = 459.$$

In CSW24, JAYHAWKERS, QUALIFICATORY, and METHOXYBENZENES give

$$9 \cdot 30 + 3 \cdot 30 + 3 \cdot 41 = 483.$$

The Supplementary Material gives legal histories for all four boards. ■

Figure 1 shows the 483-point position.

The dictionary difference is not visible from the three final words alone. Although JAYHAWKERS and QUALIFICATORY occur in both dictionaries, the required predecessors JA and QUALIFICATOR do not occur in NWL23. A record board depends on the words that existed before the record move, not merely on the words scored after it.

4. THREE TILES

Proposition 12. $M_3(\text{NWL23}) = 882$ and $M_3(\text{CSW24}) = 912$.

Proof. Proposition 8 and Table 1 find no NWL23 scoring product at 883 or above, while the largest CSW24 product at that threshold is 912. Hence the two upper bounds are 882 and 912.

Both records use the main word WHIPPERSNAPPERS, made by placing W, S, S at A15, H15, O15. Four displayed P's force two blanks, so the tripled main word scores $27(29 - 6) = 621$. In NWL23 the crosses RICKSHAW, OBJECTIVIZES, and DEMISEMIQUAVERS add $60 + 105 + 96$, for 882. In CSW24 the crosses KABELJOUW, METHOXYBENZENES, and QUALIFICATIVES add $75 + 123 + 93$, for 912. The supplementary histories use 61 and 65 tiles, so Lemma 9 supplies legal racks and draws. ■

Here the entire thirty-point difference comes from the crosswords. The final W of KABELJOUW, together with the short supporting words EE and OU, is available only in Collins.

5. FOUR TILES

Proposition 13. $M_4(\text{NWL23}) = 1130$ and $M_4(\text{CSW24}) = 1349$.

Proof. Proposition 8 and Table 1 exclude NWL23 scores of 1131 or more and CSW24 scores of 1350 or more.

For NWL23, place O, Y, O, S at A15, C15, H15, O15 to make OXYHAEMOGLOBINS. The main word scores 891. The crosses BALDACHINO, PLETHYSMOGRAPHY, JOCKO, and VENTRILOQUIZES add $54 + 26 + 54 + 105$, giving 1130. Two old H's in the column-C cross are blanks.

For CSW24, place M, B, Z, S at A15, H15, K15, O15 to make METHOXYBENZENES. It scores $27 \cdot 41 = 1107$. The crosses HYPOCHONDRIASM, JAMB, WOOTZ, and QUICKSILVERINGS add $78 + 45 + 17 + 102$, giving 1349. An old blank H accounts for the reduction from 90 to 78 in the first cross. The two histories use 84 and 72 tiles and therefore satisfy Lemma 9. ■

One product tied at 1349 reveals why reachability cannot be ignored. It contains PROVERB from H9 to H15. Since the scored word is maximal, H8 must be empty; but the opening move covers H8, and tiles are never removed. The scoring pattern is sound, yet no game can reach it.

6. FIVE TILES

Proposition 14. $M_5(\text{NWL23}) = 1401$ and $M_5(\text{CSW24}) = 1478$.

Proof. Proposition 8 excludes NWL23 scores of 1402 or more. For CSW24, the ten scoring skeletons at 1479 or more all contain maximal PROVERB from H9 to H15. Maximality forces H8 to be empty, but every one is a nonopening position and H8 has been occupied since the first move. Thus they are all impossible.

The NWL23 construction places D, L, O, Z, E at A15, E15, H15, L15, O15 to make DICHLOROBENZENE. The main word scores $27(32 + 10) = 1134$. Its five crosses add

$$\begin{aligned} \text{HYPERVENTILATED} &: 81, & \text{OVERFULFILL} &: 20, \\ \text{JOCKO} &: 54, & \text{QUARTZ} &: 34, & \text{MYXAMOEBAE} &: 78. \end{aligned}$$

for 1401.

The CSW24 construction places M, H, B, Z, S at A15, D15, H15, K15, O15 to make METHOXYBENZENES. The new H is doubled, so the main word scores $27(41 + 4) = 1215$. Its five crosses contribute $78 + 21 + 45 + 17 + 102 = 263$, giving 1478. The histories use 78 and 81 tiles. ■

7. SIX TILES Six tiles are the first case in which plausible scoring patterns survive the bag calculation and require a systematic completion argument.

Proposition 15. $M_6(\text{NWL23}) = 1723$ and $M_6(\text{CSW24}) = 1721$.

Proof. For NWL23, apply the compatible-row test to the twenty skeletons listed in Table 1. Four force the incompatible row

CO.Q . . . D . . . G . . A.

The other sixteen leave an old X at B1 with A1 and C1 unavailable before the move. It must be joined downward by a mandatory hook. None of the 165 NWL23 words of board length beginning with X matches the forced letters and leaves all rows compatible. Proposition 8 now excludes 1724.

The attaining move places O, X, P, B, Z, E at A1, B1, D1, H1, L1, O1, forming OXYPHENBUTAZONE. The main word scores 1458, and its six crosses add $69 + 11 + 34 + 57 + 31 + 63 = 265$, for 1723. Its 22 setup moves place 79 tiles; the six-tile record move brings the board total to 85, so Lemma 9 applies.

For CSW24, compatible rows and mandatory hooks leave seven skeletons. For each, the shared-bag completion enumerates every hook and every required row interval jointly, debiting their union from one physical tile multiset. None survives, so 1722 is impossible.

The CSW24 move places O, Y, P, B, Z, E at A1, C1, D1, H1, L1, O1. Again the main word is OXYPHENBUTAZONE and scores 1458. Its crosses are OPACIFYING, YARAKS, PREQUALIFIED, BRAINWASHING, ZOOGAMETES, and EJACULATIONS, worth $63 + 13 + 29 + 63 + 32 + 63 = 263$. The 25 setup moves place 87 tiles, and the record move brings the board total to 93. One player reserves the seven tiles of the last setup move, PACIFYING; the full rack-and-draw certificate verifies that neither player goes out early. ■

The international dictionary has far more playable entries, but its six-tile maximum is two points lower. The NWL23 construction relies on OPACIFICATIONS, which is absent from CSW24. Dictionaries are not nested, so their sizes do not order their records.

8. SEVEN TILES For seven tiles, the 50-point bonus is constant. We therefore state exclusion thresholds before the bonus.

Lemma 16 (Three-component connector). *If at most R new squares connect three old components, some connecting subset is the union of two simple paths from one component to the other two; neither path passes through the third component.*

Proof. Contract the old components to three terminals. Choose an inclusion-minimal connecting set of new vertices, take a spanning tree, and prune nonterminal leaves. Either one terminal lies on the path between the other two, or the tree has one branching vertex with three terminal branches. In the first case root at the middle terminal; in the second root at any terminal. Two root-to-terminal paths cover the tree, so their union uses no more than the original R new vertices. If a lifted path meets its target component early, stop there. ■

Proposition 17. $M_7(\text{NWL23}) = 1786$ and $M_7(\text{CSW24}) = 1787$.

Proof. NWL23. At pre-bingo score 1737, the compatible-row and mandatory-hook tests leave the 34 skeletons shown in Table 1. All use OXYPHENBUTAZONE across the top row. Removing the seven new letters leaves old top-row blocks that must connect downward: the board edge is above them, while the intervening new squares were empty before the turn. Enumerating every mandatory dictionary hook and every required row repair eliminates 33 skeletons. The last yields 193 reduced boards.

Each reduced board has three components, both blanks already committed, and at most six real tiles left. By Lemma 16, any completion contains one of the enumerated two-path connector cores. All three choices of root, all simple paths of the permitted length, and every multiset-respecting letter assignment are tried. For each assignment, every row and then every column is allowed to use the entire leftover multiset independently when repairing its nonwords. This relaxation considers 24,293 cores and 5,518,820 assignments. Only six pass every row, and each fails a column. Thus no pre-bingo score of 1737 is possible, so 1787 is impossible.

Lucassen's construction places O, X, P, B, N, Z, E on the seven empty squares of the top-row OXYPHENBUTAZONE. Its words contribute

$$1458 + 69 + 11 + 34 + 13 + 57 + 31 + 63 + 50 = 1786.$$

The board uses 97 tiles. A two-player certificate reserves LADDERLIKE for the second player's last setup move and follows both racks and every draw through the 24 setup moves and the record move.

CSW24. At pre-bingo score 1738, the first necessary tests leave nine skeletons. For each, the shared-bag completion enumerates every mandatory hook and row repair together and finds none. Proposition 8 therefore excludes every total of 1788 or more.

For equality, place O, X, P, B, A, Z, E at A1, B1, D1, H1, K1, L1, O1. The main word is OXYPHENBUTAZONE; the crosses are OPACIFYING, XED, PREQUALIFIED, BRAINWASHING, AMELIORATIVE, ZOOGAMETES, and EJACULATIONS. The score is

$$1458 + 63 + 11 + 30 + 63 + 17 + 32 + 63 + 50 = 1787.$$

Figure 2 shows the board. Its 27 setup moves place 89 tiles; the record move brings the board total to 96. The second player reserves the seven tiles of the last setup move, PACIFYING, then draws four tiles and does not go out. The first player holds ABEOPXZ, plays the record, and goes out. The certificate verifies every intermediate word, blank, rack, draw, and score. ■

All eight words scored by this Collins move are also in NWL23. Nevertheless, the board cannot be copied into North American play: JACULATIONS, which must exist before the final E makes EJACULATIONS, is absent from NWL23. Once again, the decisive dictionary word is a predecessor rather than a word on the final score sheet.

Propositions 12–17, together with the one- and two-tile enumeration, prove the upper bounds in Theorem 1. The displayed constructions and the complete histories in the Supplementary Material prove equality.

9. REPRODUCIBILITY The accompanying source package regenerates all fourteen finite enumerations from the two identified digital dictionaries. Independent programs verify each attaining history, including alignment, word membership, the centre-square rule, incidental words, permanent blanks, physical tile counts, racks, and draws. Small artificial dictionaries test the placement enumeration for every

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	O ₁	X ₈	Y ₄	P ₃	H ₄	E ₁	N ₁	B ₃	U ₁	T ₁	A ₁	Z ₁₀	O ₁	N ₁	E ₁
2	P ₃	E ₁	a ₀	R ₁		R ₁	O ₁	R ₁	T ₁	TL	M ₃	O ₁	O ₁	DW	J ₈
3	A ₁	D ₂	D ₂	E ₁	E ₁	m ₀	DL	A ₁	DL	R ₁	E ₁	O ₁	DW		A ₁
4	C ₃			Q ₀			D ₂	I ₁	V ₄	U ₁	L ₁	G ₂	E ₁		C ₃
5	I ₁			U ₁	DW			N ₁		B ₃	I ₁	A ₁	S ₁		U ₁
6	F ₄	TL		A ₁		TL		W ₄		TL	O ₁	M ₃		TL	L ₁
7	Y ₄		DL	L ₁			DL	A ₁	DL		R ₁	E ₁	DL		A ₁
8	I ₁			I ₁				S ₁			A ₁	T ₁			T ₁
9	N ₁		DL	F ₄			DL	H ₄	DL		T ₁	E ₁	T ₁	R ₁	I ₁
10	G ₂	TL		I ₁		TL		I ₁		TL	I ₁	S ₁		TL	O ₁
11				E ₁	DW			N ₁			V ₄				N ₁
12	DL			D ₂				G ₂			E ₁	DW			S ₁
13			DW				DL		DL				DW		
14		DW				TL				TL				DW	
15	TW			DL				TW				DL			TW

Rack A₁ B₃ E₁ O₁ P₃ X₈ Z₁₀

Figure 2. The 1787-point CSW24 move. Black tiles are newly placed; italic letters are blanks. Both blanks lie outside the words scored on the final turn.

$k = 2, \dots, 7$; further tests target transposition, boundary fragments, three-blank impossibility, shared-square penalties, and completion pruning.

The top-level driver is `scripts/reproduce_release.py`. It compiles the two enumeration programs, exports and hashes both word sets, repeats every upper-bound stage in Table 1, and then invokes separate forward verifiers for the fourteen histories. The Supplementary Material gives exact filenames and SHA-256 hashes for the one-tile and consolidated upper-bound certificates, the witness collection, the fresh-run state and verdict, and the dense NWL23 rack certificate.

The proof is computer-assisted, not formally verified in a proof assistant. Its trust base consists of the stated reductions, integer arithmetic, the published source, the compiler and interpreter, and the two version-matched dictionaries. Licensed dictionary data are identified by cryptographic hashes but are not redistributed. The Supplementary Material places each of the fourteen record boards and its complete tile-placement history on one page; passes and rack refills remain in the machine-readable certificates.

The point of the calculation is ultimately simple. A huge word and three triple-word squares are only the beginning. The same physical blank may affect two scores; an innocent-looking predecessor may belong to the wrong dictionary; and the last bridge between two parts of a board may exhaust the bag. A record needs all three layers at once: a collection of scoring words, a legal board, and a game that can actually reach it.

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